AYE
TECH HUB

• PROTASTRUCTURE • LESSON 05

Beam Design

Bending · Shear · Deflection · Reinforcement · ProtaStructure

Lesson 05 of 30 | ProtaStructure Program

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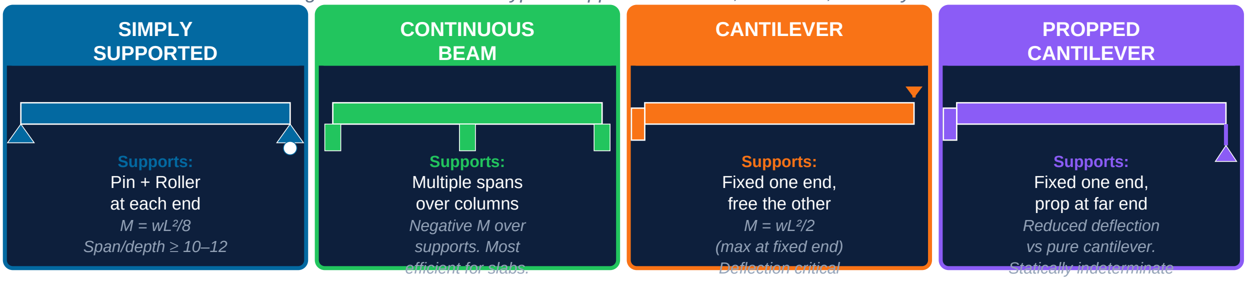
- Beam types: simply supported, continuous, cantilever, and propped
- Bending moment and shear force diagrams — how beams carry loads
- Design checks: bending, shear, deflection, and crack width (Eurocode 2)
- Beam reinforcement: top bars, bottom bars, links (stirrups), and cover
- Step-by-step beam design workflow in ProtaStructure with result interpretation

Beams — The Horizontal Backbone of Every Structure

A beam is a horizontal (or near-horizontal) structural member that carries loads primarily by bending. It transfers loads from slabs and walls down to columns and walls below. **Every floor you stand on is carried by beams.** In a reinforced concrete building, beams typically span between columns at each floor level. ProtaStructure analyses and designs beams automatically — but understanding what the software is doing is essential for interpreting results and making engineering judgements.

Figure 1 — Four Beam Types: Support Conditions, Moments, and Key Rules



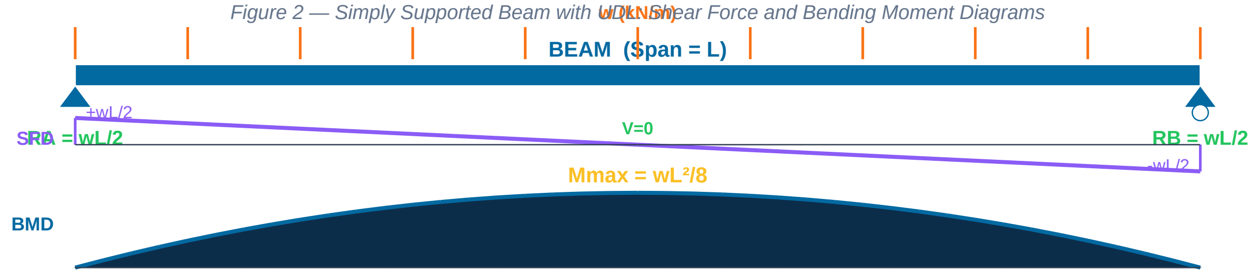
Beam Type	Structural Behaviour	Where Used	Design Note
Simply Supported	Positive bending moment throughout span. Maximum moment at midspan: $M = wL^2/8$. No moment transferred to supports.	Precast beams, bridge beams, isolated single-span beams.	Simplest to design. Least efficient — large midspan moment. Beware of deflection for long spans.
Continuous	Negative moments over internal supports, positive moments in spans. Smaller midspan moments than simply supported.	In-situ RC floor beams — the standard for multi-story buildings in ProtaStructure.	Most efficient use of material. Requires careful attention to negative moment reinforcement at supports.
Cantilever	All moment at the fixed support (max). $M = wL^2/2$ — twice the simply supported value for same span/load. Deflection increases as cube of length.	Balconies, overhanging roofs, retaining wall stems, crane beams.	Deflection governs for spans > 2m. Limit span to L/3 of back span. Always check crack width.
Propped Cantilever	Fixed at one end, simply supported at the other. Statically indeterminate — moments distributed between fixed end and span.	Transfer slabs, long balconies with mid-support column.	Intermediate efficiency between cantilever and continuous. ProtaStructure solves automatically.

Span-to-depth rules of thumb (Eurocode 2 / BS 8110): Simply supported beam: $L/d \geq 10$ (shallow) to $L/d \leq 16$. Continuous beam: L/d up to 20 (more efficient). Cantilever: $L/d \leq 7$–8 (stiffness governs). These ratios prevent excessive deflection without full calculation — use them for preliminary sizing before running ProtaStructure.

Bending Moment, Shear Force, and Reinforcement

Before ProtaStructure can design a beam, it performs structural analysis: it calculates the bending moment (M) and shear force (V) at every point along the beam under the applied loads. The reinforcement is then chosen to resist these actions.

Figure 2 — Simply Supported Beam with UDL, Shear Force and Bending Moment Diagrams



Concept	Definition	Design Implication
Bending Moment (M)	The internal force that tries to bend the beam. Measured in kNm. Positive M = beam sags (tension at bottom). Negative M = beam hogs (tension at top, over supports).	Bottom bars resist positive moment (tension). Top bars resist negative moment. Larger M → more steel or bigger section.
Shear Force (V)	The internal force that tries to slice the beam vertically. Measured in kN. Maximum at supports, zero at midspan (for UDL).	Links (stirrups) resist shear. Close link spacing needed near supports. ProtaStructure outputs link size and spacing.
UDL (Uniform Dist. Load)	Load spread evenly along the beam length: w kN/m. Comes from slab self-weight, finishes, live loads, and walls above.	For a simply supported beam: $M = wL^2/8$. Doubling the span quadruples the moment — longer beams need dramatically deeper sections.
Point Load (P)	Concentrated load at a specific location: P kN. From columns above, secondary beams, or plant/equipment.	Creates a triangular BMD. Maximum moment directly under the load: $M = P \cdot a \cdot b/L$ (at distance a from left support).
Effective Depth (d)	Distance from the top of the beam to the centroid of the tension reinforcement. d = h – cover – link diameter – bar_radius.	The lever arm for moment resistance depends on d. Never confuse h (total depth) with d (effective depth) in calculations.

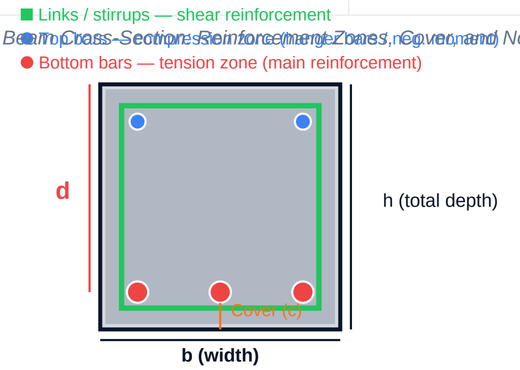


Figure 3 — RC Beam Cross-Section: Reinforcement Zones, Cover, and Notation

Concrete is strong in compression, weak in tension. Steel is strong in both. In a beam under positive bending, the top is in compression (concrete handles it) and the bottom is in tension (steel handles it). This is the fundamental principle of reinforced concrete design — every piece of reinforcement you add in ProtaStructure serves this purpose.

Beam Design in ProtaStructure — Workflow and Checks

Design Check	Eurocode 2 Clause	What ProtaStructure Verifies	Pass Condition
Bending (flexure)	EN 1992-1-1 Clause 6.1	Required tension steel As,req from Msd. Checks that provided As ≥ As,req. Also checks compression steel if needed.	As,prov ≥ As,req x/d ≤ 0.45 (ductility) Minimum As: 0.0013·b·d
Shear	EN 1992-1-1 Clause 6.2	Design shear VEd vs concrete shear capacity VRdc. If VEd > VRdc: provides shear links (stirrups).	VRdc ≥ VEd (no links needed), or Link spacing s ≤ min(0.75d, 300mm) VRdmax ≥ VEd (no crushing)
Deflection	EN 1992-1-1 Clause 7.4	Span-to-effective-depth ratio check. Compares actual L/d vs allowable L/d (modified for steel and concrete stress).	Actual L/d ≤ Allowable L/d Or computed deflection ≤ L/250 (or L/500 for sensitive finishes)
Crack Width	EN 1992-1-1 Clause 7.3	Controls crack width under service loads. Governed by bar spacing and bar diameter.	wk ≤ 0.3 mm (XC1 exposure class) wk ≤ 0.2 mm (aggressive exposure) Max bar spacing ≤ 200mm
Minimum / Maximum Steel	EN 1992-1-1 Clause 9.2	Ensures beam is not under- or over-reinforced. Prevents brittle failure and ensures ductility.	As,min = 0.26·fctm/fyk·b·d ≥ 0.0013·b·d As,max = 0.04·Ac (4% of gross area)

Step-by-Step Beam Design in ProtaStructure

Step	Action in ProtaStructure	What to Check
1. Model beam	Draw beam between column nodes on floor level. Set section size (e.g. 300×600mm) and material (C25/30, B500B).	Check span lengths are correct. Ensure beam is connected to columns at both ends.
2. Define loads	Assign slab loads (self-weight + finishes + live load) to beam via tributary area or direct UDL input.	Verify load combinations include 1.35G + 1.5Q (Eurocode fundamental combination for ULS).
3. Run analysis	Analysis → Run All. ProtaStructure performs FEM analysis and generates BMD and SFD for all load cases.	Check BMD shape is as expected. Red/orange beams in model = overstressed.
4. Design beam	Design → Beams → Design All (or select specific beams). ProtaStructure auto-selects reinforcement to satisfy all checks.	Review design ratio (DR): DR ≤ 1.0 = OK. DR > 1.0 = increase section or concrete grade.
5. Review results	Open Beam Design Report for each beam. Check: As,req, As,prov, shear links, deflection ratio, crack width.	Compare provided steel to required steel. Ensure deflection ratio passes. Check link spacing near supports.

6. Optimise	Adjust beam depth or width if design ratio is very low (< 0.5) or too high (> 0.9). Re-run design after section changes.	Target design ratio: 0.6–0.85 for economy. Use standard section sizes: 200, 250, 300, 400mm widths.

Common ProtaStructure Beam Warning	Meaning	How to Fix
Design ratio > 1.0 (overstressed in bending)	Required steel exceeds maximum allowable — section too small or load too high	Increase beam depth (most effective — $M \propto d^2$). Or increase concrete grade (C30 → C35). Widen beam only as last resort.
Shear check fails (V_{Ed} > V_{Rdmax})	Concrete section is too small even with maximum links — compression strut crushing governs	Increase beam width (shear resistance $\propto b \cdot d$). Increasing depth also helps but less effective for shear.
Deflection ratio fails (L/d > allowable)	Beam will deflect excessively under service loads — risk of cracked finishes or damaged partitions	Increase beam depth. Add compression steel. Reduce span or add intermediate support.
Insufficient cover warning	Cover set in model is less than minimum for exposure class	Open Section Properties → adjust cover. Use 25mm for XC1 (dry), 35mm for XC2/XC3, 45mm for XC4.

What comes next — PROTA-006: Column Design — how columns carry axial loads and bending moments, slenderness effects, biaxial bending, column reinforcement patterns and minimum steel, and the complete column design and optimisation workflow in ProtaStructure.